

22CH203 & Engineering Chemistry II

1. * The reaction with lower activation energy (low peak) is faster because the reaction occurs more easily.
* The reaction with larger energy difference is more stable because the products have lower energy.
* Thus, low activation energy increases speed, while large energy difference increases stability
2. * The reaction releases a large amount of energy rapidly, so it is hazardous and hard to control
* $\therefore$ It needs strict safety precautions and controlled conditions
3. * All substances are in same liquid, so reaction happens faster
* Catalyst mixes completely and increases reaction efficiency
* Intermediate substance forms temporarily, showing the reaction occurs step by step before final product form
4. * After a certain limit, adding more catalyst does not increase the reaction rate because reactant concentration becomes the limiting factor
* The system reaches saturation, so the reaction rate becomes almost constant
5. * Low activation energy process starts easily and is safer but gives less energy output
* High activation energy process needs more initial energy, but produces higher energy and better productivity
* For large-scale continuous production, the high energy output process is preferred if safety and cost are properly managed

6. * The reaction is an exothermic reaction because the products have lower energy than the reactants.
* The single peak ~~character~~ represents the activation energy needed for the reaction to occur.
7. * Solvent polarity changes the reaction speed by affecting reactant-catalyst interaction.
* Higher catalyst concentration increases reaction rate.
* Solvent and catalyst together decide which product is formed more.
8. * The homogeneous catalytic system gives high reaction efficiency and better product yield.
* However, catalyst recovery is difficult and increases process cost and complexity.
9. * The forward reaction has lower activation energy, so it occurs faster and more easily.
* The reverse reaction has higher activation energy, so it proceeds more slowly.
* Because of this difference, the reaction initially favours product formation.
* At equilibrium, both forward and reverse reactions occur at equal rates, but concentrations remain constant.
* Lower energy products are more stable, making the reaction more feasible and thermodynamically favourable.
10. * The powdered catalyst has a large surface area compared to large solid chunks.
* Most surface area provides more active sites for reactant molecules to absorb and react.

* This increases the frequency of effective collisions and speed up the reaction.

* In large chunks, only a small outer surface is available, so the reaction rate is slower.

* $\therefore$ catalytic efficiency increases with increase in surface area and availability of active sites.

11. * The energy barrier must be crossed before reactants form products.

* Even though the reaction is thermodynamically favorable, high activation energy makes the reaction slow at room temperature.

* The peak in the energy profile represents the high-energy intermediate state.

* Increasing temperature gives more energy to cross the barrier.

* $\therefore$ the no. of effective collisions increases and the reaction rate becomes faster at high temperature.

12. * In heterogeneous catalysis, reactants first adsorb on the catalytic surface, then react and finally products are formed.

* Increased surface roughness gives more active sites, so adsorption and reaction rate increases.

* Moderate temperature increases collision frequency and reaction rate.

* Excess temperature causes side reactions and catalyst deactivation, reducing efficiency.

* The surface reaction at active sites controls the overall reaction.

13. * The enzyme is substrate-specific, and forms only the required product.
* Small amounts of enzyme give a fast reaction showing high efficiency.
* The enzyme is not consumed and can be reused repeatedly.
* It increases reaction rate but does not change equilibrium position.
* High selectivity, reusability, and efficiency make enzymes useful in industries.
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14. * Substance increasing the rate is catalyst, the one decreasing the rate is an inhibitor.
* Catalysts lower activation energy and speed up the reaction.
* Inhibitor slows the reaction by interfering with the mechanism.
* They change only the reaction rate, not the equilibrium composition.
* Impurities affect catalyst performance, so careful control is needed.
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15. * The iron catalyst provides an alternative reaction pathway with lower activation energy.
* Lower activation energy increases effective collisions and speeds up the reaction.
* This catalyst increases reaction speed without changing temperature or pressure.
* The catalyst remains chemically unchanged after the reaction and can be reused.
* The lower peak in energy profile shows reduced activation energy in the catalysed reaction.

16. * Homogenous catalysis (same phase) gives fast, uniform reaction, but catalyst separation is difficult.
- * Heterogenous catalysis (solid catalyst + gas reactants) occurs on solid surface and is easy to separate and reuse.
- * Homogenous system is more efficient but less practical for large-scale production.
- * Heterogenous catalysis is slightly slower but more economical and scalable.
- * Hence, heterogenous catalysis is preferred for industrial use.
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